

Capstone Project Write-Up
Bridging Data, Place, and Purpose:
A Grantmaking Equity Analysis for TFFF

Aiyana Brown

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1 Title, Team, and Affiliations

Project title: *Bridging Data, Place, and Purpose: A Grantmaking Equity Analysis for TFFF*

Aiyana Brown | Owner | acbrow2@willamette.edu |

Peer Stakeholder POs: *Summer Tucker, Sophia Rabbanian, Tiffany Truong*

Repository: <https://github.com/AiyanaBrown/MSDS-Capstone-Project>

2 Abstract

Nonprofit organizations often hold community data across disconnected systems, and The Ford Family Foundation (TFFF) was no exception. This made it difficult to assess whether grant and

scholarship dollars were reaching the communities most in need. This project combined TFFF's grant, scholarship, and Oregon by the Numbers (OBTN) need-indicator data into a single analytical pipeline to measure how well funding under the 2021-2025 strategic plan aligned with community needs across rural Oregon and Siskiyou County, California.

After cleaning, imputing, and joining the data by county and year, alignment between funding and need was tested through Spearman correlation analysis, a composite need index built from TFFF's three strategic pillars (Strong Starts, Bright Futures, Thriving Communities), and LASSO-selected Poisson and Gamma regression models. Across \$176 million in total funding (2,700+ grants and 1,400+ scholarships), grant dollars were positively correlated with community need, especially foster care rates and high school graduation rates, and internally consistent with TFFF's mission. Scholarship dollars, however, were weakly and sometimes negatively correlated with need, concentrating more in higher-population, urban counties rather than rural communities. A Douglas County deep-dive showed funding aligned more closely with need locally than statewide, and need indices have shifted meaningfully from pre and post-pandemic conditions—most notably in rising financial hardship, unemployment, and worsening health indicators, which is a signal for the next strategic plan. A reusable R pipeline and interactive dashboard were built alongside the analysis so TFFF can continue tracking funding-need alignment.

Because TFFF's 2021-2025 strategic plan was itself built considering OBTN need indicators, funding was intended to be responsive to community needs by design. This analysis measures the degree to which realized funding reflects the intended relationship, rather than establishing causality. Results should still be read alongside limitations, including imputed values, geographic dollar-splitting, non-population normalized values, and a single-year modeling snapshot (36 counties) required by OBTN methodology changes across editions.

3 Problem and Research Questions

Problem Statement

The Ford Family Foundation (TFFF) has invested millions of dollars in grants and scholarships to support rural Oregon, and it has demonstrated a clear commitment to understanding community needs by collecting data like Oregon by the Numbers and Oregon Voices. However, these sources lived in different places. Grant records, scholarship data, funding requests, industry statistics, and demographic/geographic data were stored separately as disconnected Excel files without a built-in interface to combine them. Siloing inherently related data makes it difficult to see the full picture: whether funding patterns align with documented needs, how those needs are shifting over time, and where future investments might have the greatest impact. While the challenges facing rural Oregon are well-known, there is limited research that systematically maps philanthropic funding distributions against community needs at the geographic level with a focus on building connection and opportunity. My project aims to help fill this gap and better understand where resources can do the most good. Creating a more cohesive analytical framework gives the Foundation a stronger

quantitative grounding to understand what's working, communicate impact clearly, and plan strategically for rural Oregonians moving forward.

Personal Motivation

I am a Ford Scholar, coming from a low-income family in a rural community. The Foundation's generosity has allowed me to be the first in my family to attend university, and now I am getting my Master's degree with their financial support. Their help has enabled me to overcome incredible adversity, find a career path that I love, and build a better life for myself. My goal through this project is to help the Foundation reach more people like me by providing scholarships and community support to the people and organizations who wouldn't otherwise have that opportunity.

Mission

I will design and implement an analytical framework for the Ford Family Foundation that integrates grant records, scholarship data, and Oregon community statistics to evaluate geographic equity under the previous strategic plan, analyze gaps and successes, and offer recommendations for more responsive funding decisions for rural Oregon communities. The final deliverable will be a cohesive, end-to-end data pipeline feeding into an interactive dashboard interface to help TFFF staff navigate, visualize, and interpret community data for decision-making.

Primary Research Question: To what extent did TFFF's grants and scholarships under the previous strategic plan align with the needs of rural Oregon communities, and where do geographic gaps or disparities suggest priorities for the next strategic plan?

Secondary Research Question: Can I improve the current data infrastructure (connecting all relevant data in one place) to support more integrated, data-informed grantmaking going forward?

Changes since M1 proposal: *A second version of M1 proposal was submitted. No additional changes have been made.*

4 Impact and Stakeholders

To understand the impact of this project, it is important to be familiar with the Ford Family Foundation's work. TFFF is a nonprofit serving rural Oregon and Siskiyou County, California, with a mission to create the conditions where children have the family, educational, and community support they need to thrive. The Foundation's work is rooted in a commitment to equity and access in rural communities. Some of their work includes:

- Investing in individuals through scholarships, training, and leadership development opportunities
- Supporting organizations that strengthen social and economic conditions for children and families through grantmaking
- Equipping rural leaders and policymakers with tools and resources to engage with and shape the systems that affect their communities

- Leveraging the Foundation's own research and influence to bring attention to rural issues and encourage statewide policies that benefit such counties

Project Impact

The Ford Family Foundation and all of its stakeholders will benefit from this analysis and data pipeline. It will allow the non-profit organization to better serve Oregon communities by:

- Improving equity in scholarship access and college attainment
- Aligning responsive and strategic grant funding with real community needs
- Advising leadership in creating the next strategic plan

This work supports rural Oregon children, families, students, and businesses, and it may inform policymakers making program and funding decisions across the state.

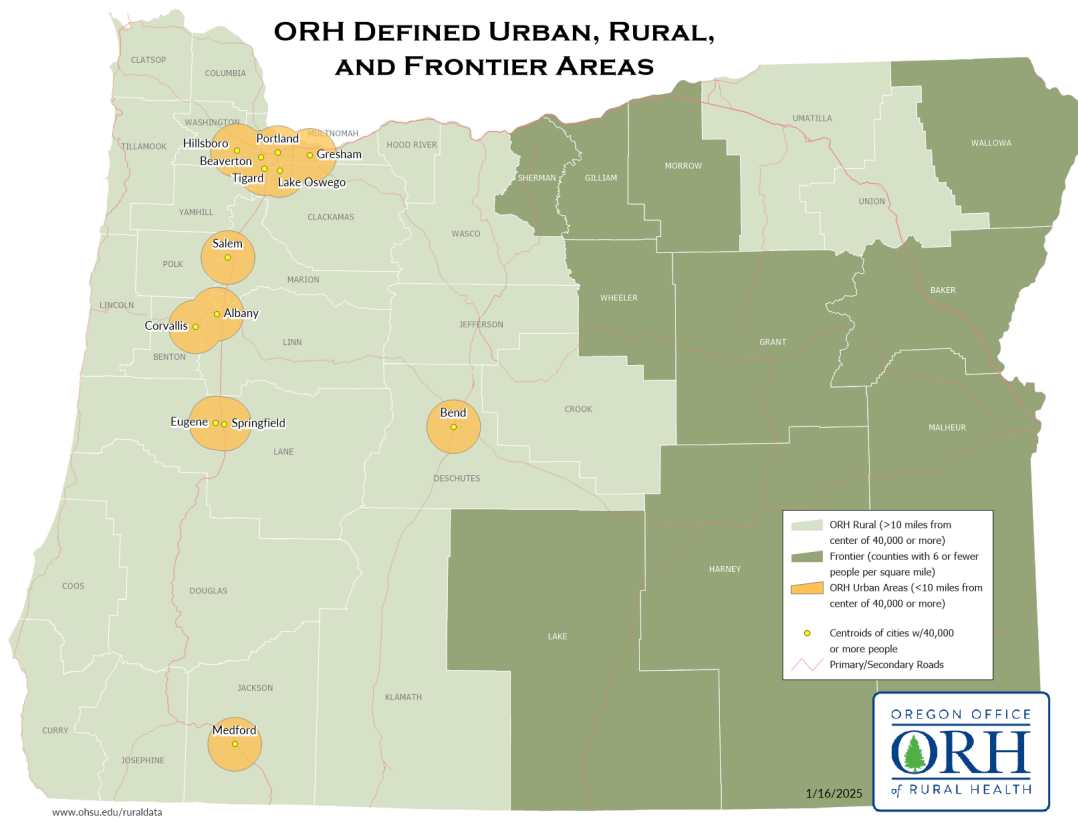
5 Related Work and Background

As an internal analysis for a single foundation, this project draws on two bodies of prior work: research on equitable grantmaking practice, and the need data TFFF already collects.

In a survey-based study conducted with community foundation executive directors across the United States, Azevedo and Bell (2025) found that community foundations organizing funding decisions around explicit mission and strategic goals achieve a stronger perceived equity in decision-making. This project seeks to extend this logic: rather than surveying perceived equity, it measures whether TFFF's actual grant and scholarship dollars align with social need indicators tied to its own strategic pillars.

Each year, the Ford Family Foundation has published an updated release of *Oregon By The Numbers*. This report details key need indicators, community statistics, and industry facts about each county in Oregon. Data from the 2021-2025 OBTN reports are sourced from the community need measure used in this analysis.

The OHSU Oregon Office of Rural Health has defined and listed rural counties in Oregon, which can be referenced when analyzing fund distributions. Rural areas are 10+ miles from a population center of 40,000+, and frontier counties have six or fewer people per square mile; these classifications were used throughout the geographic analysis. A map from the OHSU website below shows the population distribution across Oregon.



Map reference: [19] OHSU

Methodological choices (Spearman correlation for skewed, outlier-prone data [7, 14, 15]; per-type IQR fencing; bivariate multicollinearity analysis [6]; need indices [2]; quasi-Poisson/Gamma models for overdispersed dollar data [7, 9, 10, 11]; LASSO with bootstrap resampling for variable selection under a small sample [4, 5]) were informed by prior applications to similarly structured data. See the Methods section for reasoning and References for full citations.

6 Data and Data Engineering

All datasets are Excel files spanning 2021–2025 unless otherwise noted.

Source	Description	Access	Volume	Approval
Grant Records	Strategic and responsive grant records, including the ~\$25M responsive grant pool	Private	~2,000 records 20+ columns	Approved

Scholarship Records	Student award tracking and community college distribution data	Private	~2,000 records, 10–15 columns	Approved
Oregon by the Numbers	5-year workbook covering 47 community-level measures (109 raw columns) across all 36 Oregon counties, including demographics, industries, and key challenges	Public	47 measures (109 raw columns) × 36 counties	Approved

Notes: Oregon Voices data can be used as supplementary information, not in the full ingestion process. This is due to PII within the dataset per our data-sharing agreement. The county will serve as the primary unit of analysis. Where applicable, the project will account for pass-through grants (e.g., funds distributed to Portland organizations like CASA for later rural deployment).

Public data web access:

1. [Oregon by the Numbers | The Ford Family Foundation](#)
2. [Oregon Voices](#) (Supplementary)

Data Engineering Plan

All data is external and provided as Excel files.

data/

raw/ original datasets from TFFF (gitignore)

external/ analysis-ready snapshots of private data (gitignore)

processed/ cleaned, analysis-ready snapshots

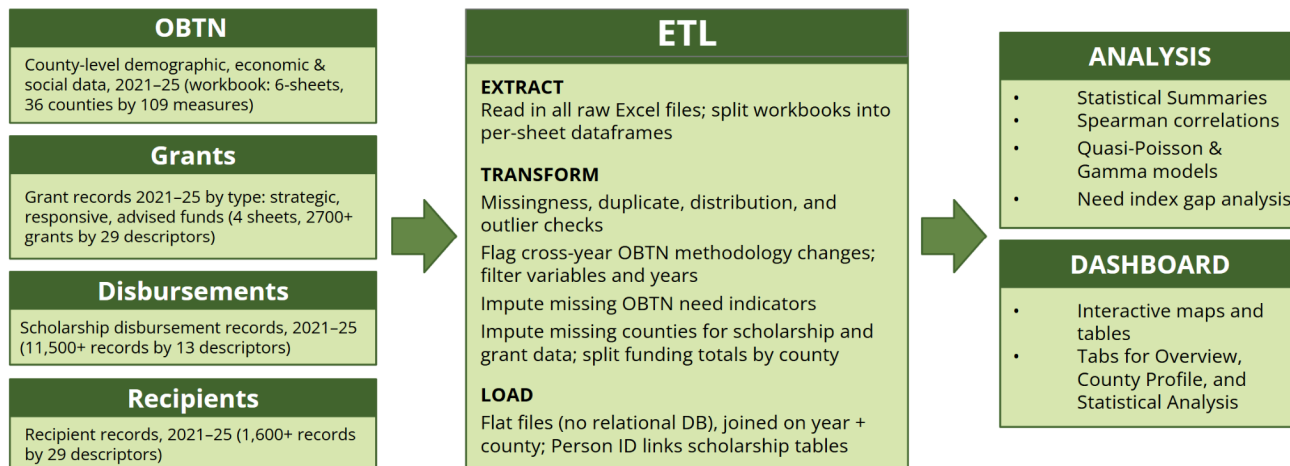
README.md describes each dataset: source, license, date, size

SCHEMA.md describes processed dataset schemas once they stabilize

Engineering Decisions

Due to scope and reproducibility concerns, I kept all data in a flat file. Building a full, reproducible data lake with ingestion scripts, along with analysis and dashboards, is outside the scope of a single-person project. A key factor guiding this design choice is practicality for TFFF. If I designed 3NF-normalized tables in a relational database, the Foundation would have to repeat that process for any new data collected. This would require a data team member to dedicate significant time to maintaining the database, and there isn't a framework in place for that yet. While normalization might be more efficient for updating data and minimizing storage, someone would still need to manually build new tables, ensure data quality, and ingest new data each time. That's a substantial ask for a team without dedicated support. Realistically, what will be most useful to TFFF is the solution that offers the greatest benefit for the least ongoing maintenance effort. This is why I've decided to keep the data in one place: I'm prioritizing future usefulness over demonstrating my ability to build normalized schemas.

Ingestion Process: Data cleaning, feature selection, dashboards, and visualization in R



7 Ethics and Responsible Use

Ethics & Data Agreement

This project operates under a Data Use and Confidentiality Agreement (effective June 17, 2026) between The Ford Family Foundation (TFFF) and the student researcher, governing access to OBTN data, Strategic and Responsive Grants data, and Scholarship/Disbursements data (2021–2025).

1. **Scope of use:** Data may be used solely for the capstone Project. It cannot be repurposed, commercialized, or published beyond academic requirements without TFFF's prior written consent.
2. **Confidentiality:** All non-public data provided by TFFF (including grantee records, financial information, and any derived summaries or analyses) is classified as Confidential Information, regardless of whether it's marked as such. This applies to outputs as well as the raw data itself.
3. **Disclosure limits:** Confidential Information cannot be shared with third parties without TFFF's written consent. Sharing with the faculty advisor and thesis committee is allowed, provided they're informed of the confidential nature and agree to honor it.
4. **Reproduction:** Copying or reproducing confidential Information is limited to what's reasonably necessary for completing the Project.
5. **Retention and destruction:** Upon TFFF's request or Project completion, all confidential information must be returned or destroyed, unless the student's institution requires retention for academic purposes, in which case retained materials remain subject to the same confidentiality terms.
6. **Ownership:** TFFF retains all rights to the underlying data. The student owns the original written work product, but any portion derived from Confidential Information can't be published or distributed beyond academic requirements without TFFF's consent.

7. **Governing terms:** Confidentiality, non-disparagement, and IP provisions survive the end of the agreement, even after the Project is submitted or the agreement is otherwise terminated.

PII Handling Note

To comply with the confidentiality terms of the TFFF Data Use Agreement, several PII elements are handled with additional care. People IDs are not displayed in any output and are used only internally to link records across datasets. Address data is aggregated to the county level rather than reported at the individual level, preventing identification of specific recipients. Grant and scholarship details, including specific dollar amounts, will be shared or published only after obtaining explicit permission from TFFF on a case-by-case basis as the analysis progresses, consistent with the agreement's requirement that confidential information not be disclosed beyond academic requirements without prior written consent. Kasi Allen approved the use of funding totals for this capstone project in August 2026.

Non-causal framing: This analysis will analyze correlations between variables. This does not establish causality. While it may signal influential factors, further research and domain context are needed before reaching concrete conclusions.

8 Methods and Evaluation

8.1 Pre-Analysis

I ran initial missingness and distribution analysis to understand how skewness and outliers may affect the project. I identified missingness in necessary columns, such as county and a few OBTN need indicators, which were flagged for imputation during data cleaning. I detected outliers using IQR fences computed per Grant Type rather than pooled across all grants. Strategic, Responsive, and Advised Fund grants operate at very different dollar scales (median award ~\$28K, ~\$10K, and ~\$92.5K respectively). Using a single quantile grouping over-flagged 357 records (12.9% of grants, holding 67.2% of total dollars), which was disproportionately concentrated in the higher-scale grant types rather than reflecting actual outlier awards. Per-grant type outlier fencing corrected this.

Funding Outliers: After examining the distribution of the grant and scholarship data, I noted outliers for further investigation rather than removing them, as they are all valid data points that trace back to real dollars given by the foundation. This is discussed further in section 8.3.

8.2 Data Cleaning

Deduping

Data cleaning began with deduplication: grants and scholarships were deduplicated, and grant requests were filtered down to those that were actually funded. Scholarships were filtered for uniqueness across recipient and scholarship type (e.g., the same recipient may receive an undergraduate and graduate award; this will count as two separate scholarships).

County-Matches

Both grant and scholarship datasets had significant missingness in columns tracking funding by county; these values were ‘imputed’ for analysis. County attribution required different logic for grants and scholarships, since the two datasets carried different geographic fields. For grants, I parsed counties from the Geographic Focus column first, then the Counties column, and, if still missing, fell back to the organization's headquarters zip code, restricted to Oregon and Siskiyou County, CA. Statewide and national-scope grants were left unresolved at this stage and handled separately (see Splitting the Dollars, below). For scholarships, original_county came from an existing field in the data, backfilled with a zip-code reverse lookup; six zip codes required manual entry by referencing the city listed because they were not in the zip code reference table. For scholarships, I derived a parallel current_county field, representing the recipient's current residence, but original_county (where the recipient is originally from) is the field used in the equity analysis, since that best matches the question the OBTN and grants maps are answering.

Imputation

Finally, I imputed missing OBTN values using the full 2021-2025 dataset. For each variable, if the value was null, it was imputed using a 3-step fallback: first, the mean of other available years for that same county; second, if no other years existed, the mean of population-peer counties within a +/-5,000 population window; and, as a last resort, the statewide mean across all counties and years. This was done to support the modeling downstream and preserve the validity of the results.

Splitting the Dollars

Grant dollars were split evenly among the counties they impacted, per the request of the internal stakeholder at TFFF. If no impacted counties were listed, the default was assigned to the location of the grant recipient's headquarters. Statewide and national-scope grants were excluded from this split entirely, rather than divided evenly across all counties in the state. Spreading them across counties would have created false uniformity in what is fundamentally an equity analysis, relying on assumptions that each county received a share of funding that may not match actual circumstances.

Joining Data

With county attribution resolved, I joined the remaining datasets: scholarship distribution and recipient data were joined on People ID, and all datasets were connected by county and year. Race and age/sex data were pivoted and cleaned for a later rejoin. Though the analysis ultimately focused on dollars and need indicators, race and age/sex data are cleaned and ready to use for future analysis.

8.3 Statewide Analysis

All correlation, need index, and modeling work in this section compares actual funding given (2021-2025) against a fixed 2021 OBTN snapshot, the edition that informed TFFF's original strategic plan. The framing is plan versus reality: did funding match the need TFFF saw when it created its strategy? Later, in the Douglas County profile, I compare that 2021 snapshot against the most recent

(2025) edition to show what has changed since the foundation last used OBTN data to guide planning.

Exploratory Data Analysis (EDA)

EDA explored top funding areas, top grant recipients, and overall funding distributions. Because grants heavily favor Douglas County, maps use quantile-based coloring rather than a continuous scale, giving a broader view of relative distribution rather than letting one outlier dominate the visuals. Given how much Douglas stood out, I ran a validation check to rule out attribution error (e.g., grants misassigned to Douglas via the headquarters-zip fallback). The check confirmed the concentration is real, not a data quality issue.

Distributions

All distributions were checked for normality before analysis. Scholarship and grant data were heavily right-skewed, as expected for funding data, and log functions were applied. Most of the 29 selected OBTN measures were roughly normal, but several showed outliers or non-normal shapes; the effect of this was tested via skewness, kurtosis, and comparison of correlation coefficients.

Normalization

All OBTN values were z-scored (mean 0, standard deviation 1) to make variables measured in different units comparable ([2] Remington et al.). Z-scoring doesn't change the Spearman results below (rank correlation is invariant to monotonic transformations), nor does it affect which variables LASSO selects, since glmnet standardizes predictors internally regardless of original scale ([22] glmnet documentation). It was retained as a prerequisite for the Need Index; combining variables measured in dollars, percentages, counts, and rates into one composite score requires a common scale first.

Correlations (Spearman Coefficients)

Correlations measured alignment between funding and community need. Testing both Spearman and Pearson (with a log transformation of total dollars for Pearson) showed meaningfully different coefficients, confirming that outliers and non-normal distributions in the OBTN data would bias Pearson estimates. Spearman was used throughout for this reason ([7] Rovetta, [14] Laerd Statistics, and [15] University of Virginia Library). Dollar amounts and counts produced no meaningful difference in results, so dollar amounts were used for all analyses. Correlation analysis was run at three levels: scholarship dollars/counts, grant dollars/counts, and all dollars combined. Strength of Correlation uses a standard convention: $|r| < 0.30$ Weak, $0.30-0.69$ Moderate, ≥ 0.70 Strong (Akoglu [13] and Mukaka [8]). This threshold was applied consistently across the Grants, Scholarships, and Overall Funding tables.

Need Index (Z-Score Composites)

The need index compares counties statewide and correlates funding directly to need; it also serves as an internal prioritization tool. Its categories and pillars come directly from TFFF's mission: Strong Starts (family), Bright Futures (education), and Thriving Communities (community). Each OBTN variable is z-scored and direction-corrected so higher always means more need, averaged within each

category (family, education, finance, health, community, geography), then grouped and averaged into the three pillars, and finally averaged into one composite index – giving each pillar equal weight regardless of how many variables feed into it ([2] Remington et al.).

Modelling

To validate the correlations above, a model was built with total TFFF dollars as the response and community need indicators as explanatory variables. The goal of this modeling was not to predict funding precisely, but to extract which factors were most influential. Because dollar amounts are non-negative, continuous, and right-skewed at the county level, a log-link Generalized Linear Model (GLM) was used instead of OLS ([9] Veazie, [10] Manning, and [11] Polgreen & Brooks). Several members of the log-link GLM family were tested to find the best match for the funding data's shape.

Multicollinearity Analysis. An initial bare-bones model revealed multicollinearity, so a bivariate Spearman collinearity analysis (threshold $|r| > 0.7$) was run on candidate predictors ([6] Vatcheva et al.). Features derived from funding totals (dollar subtotals, counts, medians) were excluded beforehand, since these are components of the response variable rather than independent measures of need. A second pass at a stricter 0.6 threshold reflected how often OBTN variables act as proxies for the same underlying statistic. Flagged variables were removed, and the remaining set was narrowed to 2-3 per category (education, finance, health, etc.) using domain knowledge and stakeholder input, before LASSO identified the most influential variables.

LASSO Variable Selection. Variable selection was performed using LASSO leave-one-out CV and bootstrapping ([4] Laurin et al. and [5] Mannan). To select an interpretable predictor set from a small sample (36 counties), I applied regularized LASSO with L1 regularization under Poisson and Gamma log links, using Leave-One-Out Cross-Validation (LOOCV) to maximize training data per fold and Poisson deviance to evaluate fit. The 1-standard-error rule (λ_{1se}) applied a conservative penalty that favored parsimony and reduced the risk of overfitting with more predictors than observations; Poisson and Gamma links yielded the same result. To validate beyond a single run, I then performed bootstrap resampling: 1,000 resampled datasets, each fit with 10-fold (not leave-one-out) cross-validated Poisson LASSO, since duplicate rows from resampling can let LOOCV leak information between folds. Variables selected in at least 70% of resamples were treated as stable "core drivers" and carried into the models below. Variables selected as most influential by the LASSO bootstrap method were used for all downstream inference. No test/train split was performed on the underlying data: with only 36 rows, every observation was necessary to retain, and cross-year validation wasn't possible due to OBTN methodology changes between editions.

All models were used to measure the strength and direction of influential need indicators; results are preliminary and should be validated further before drawing final conclusions.

Quasi-Poisson Model. This model was chosen for its non-negative, right-skewed value assumption and overdispersion parameters that match data distributions ([3] Ver Hoef & Boveng). Results showed some significance, though overdispersion remained high.

Gamma Model. A Gamma GLM better matches the variance structure of right-skewed, continuous dollar data with variance proportional to the square of the mean ([9] Veazie et al. and [10] Manning et al.). Fitting LASSO under a Gamma log-link is computationally intensive and prone to convergence instability under repeated bootstrapping, so Poisson was used for variable selection and the same bootstrap-selected variables were refit under a Gamma link for final inference.

8.4 Douglas County Profile

The Douglas County Profile used the same EDA approach as the statewide analysis, but filtered to Douglas County. Index scores compare Douglas's top areas of need in 2021 (for strategic alignment) against 2025 (for future planning). Note: Child Poverty and Financial Hardship changed OBTN methodology during this window (2025 and 2024, respectively), so any shift in those two variables may reflect the redefinition rather than an actual change in need.

8.5 Dashboard

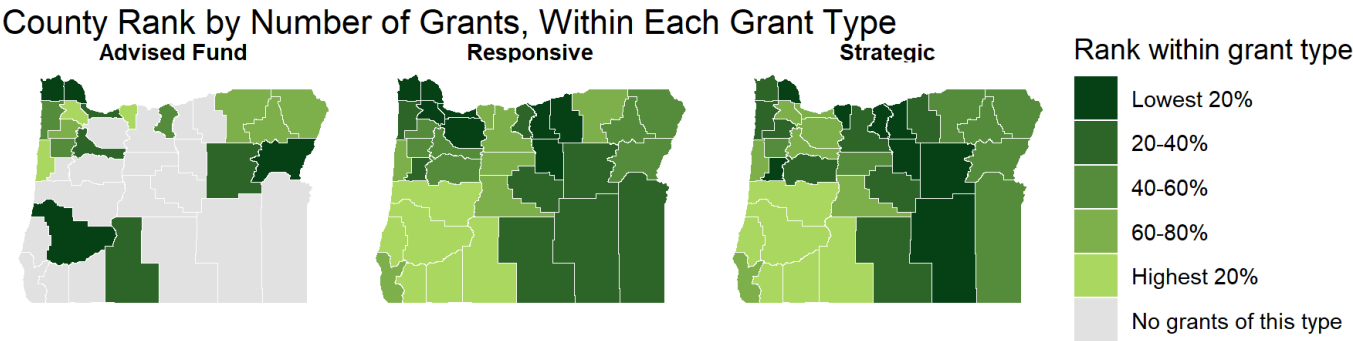
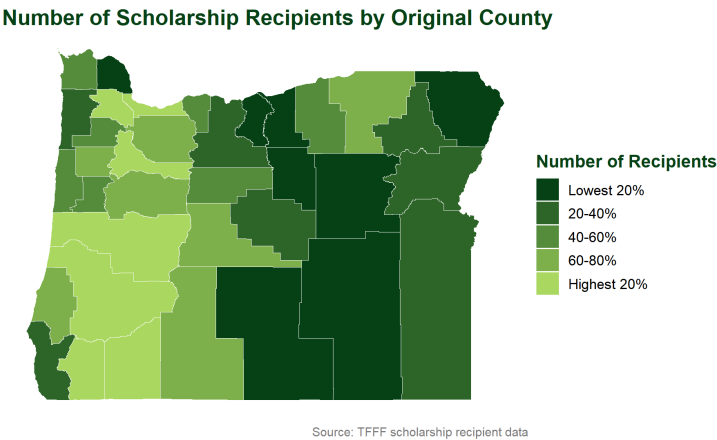
The dashboard communicates these findings and serves as a visual analysis tool, letting TFFF generate a county profile for any Oregon county. It has two tabs: a statewide overview and county profiles with interactive maps and tables for scholarship, grant, and OBTN data.

9 Results and Visualizations

9.1 Statewide Analysis

1. How much funding went toward scholarships and grants over the previous strategic plan? To what counties have those dollars and scholarships gone?

Metric <chr>	Amount <chr>
Total Scholarship Dollars	\$45,439,162
Number of Scholarships	1,431
Median Scholarship	\$26,710
Total Grant Dollars	\$130,681,193
Number of Grants	2,701
Median Grant	\$13,000
Total Dollars Given	\$176,120,355



Overall Funding: The foundation gave over \$176 million to Oregon communities during the 2021-2025 strategic plan. This consisted of 1,400+ scholarships and 2,700+ grants. On average, each scholarship recipient received ~\$27,000 and each grant recipient received \$13,000.

Grants: Advised fund grants focus on the Northwest and Northeast. Domain experts report dollars allocated to Eastern Oregon supported wildfire prevention and firefighting efforts. Strategic and responsive grants were largely concentrated in southern Oregon, although combined they do fund most rural counties in Eastern Oregon. The overall distribution indicates that responsive grants reach rural communities more than strategic grants, revealing room for greater alignment under the next strategic plan.

Scholarships: Similar to grants, the majority of scholarships (100+ recipients) were concentrated in Douglas County and Southern Oregon. All counties have at least one scholarship recipient. Many rural counties have been prioritized; however, multiple rural counties in Eastern Oregon only received 1-10 scholarships.

Q: To what organizations has the \$176 million gone?

Organization <chr>	Organization Type <chr>	Total Dollars <chr>
Oregon Community Foundation	Foundation (Peer Group)	\$10,996,881
Douglas Education Service District	Education	\$3,142,420
Protect Our Children, a project of Tides Center	Health & Human Services	\$2,613,000
Nonprofit Association of Oregon	Other	\$2,547,000
University of Oregon	Education	\$2,350,245
The Campbell Institute	Education	\$2,105,000
Rural Development Initiatives, Inc.	Civic & Community	\$1,870,500
Umpqua Community College	Education	\$1,854,679
Portland State University	Education	\$1,695,790
Foundations for A Better Oregon	Other	\$1,655,000

Findings: The top organizations are largely education-based, which aligns with TFFF pillar *Bright Futures*.

Top organizations of note are:

- 1) The Oregon Community Foundation (OCF). This organization was an outlier and received significantly more than other recipients (~\$11M). The OCF is a pass-through organization used for advised funds, which allows for flexibility to respond to urgent community needs in real time, rather than being tied to annual budget cycles. As mentioned earlier, many of these grants represent efforts to combat wildfires in Eastern Oregon. After completing core analyses, I re-ran all statistical methods without including grant data from the Oregon Community Foundation to gauge the outlier's true effect on correlations and models. There was no significant change in any results, and relationships between need indicators and funding stayed the same. This revealed that while the organization operates fundamentally differently than others, its alignment with community needs mirrors overall performance in TFFF.

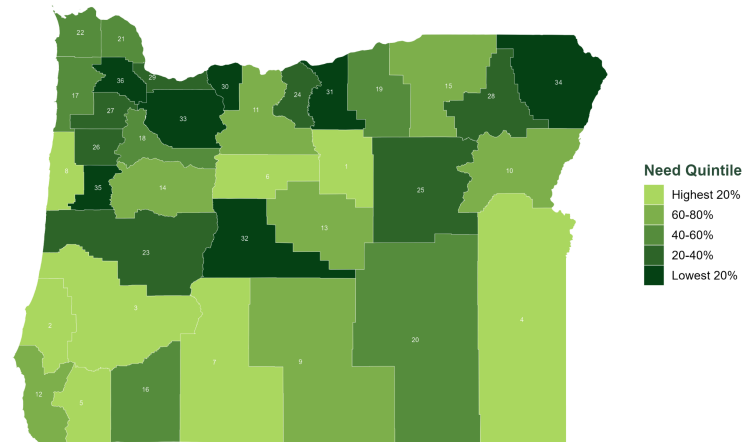
- 2) The Tides Center. The Protect Our Children project from the Tides Center was a statewide child sexual abuse awareness and prevention program. This was an internal priority under the 2021-2025 strategic plan, and it is clear that the dollars given matched intention.

Q: Based on the Index created using OBTN indicators and TFFF pillars, where is the need in Oregon? How does county need compare to total funding under the 2021-2025 strategic plan?

Geography	Need Index Score	Comparative Need Rank
Wheeler	0.718	1
Coos	0.640	2
Douglas	0.478	3
Malheur	0.458	4
Josephine	0.390	5
Jefferson	0.360	6
Lincoln	0.332	7
Klamath	0.326	8
Lake	0.307	9
Baker	0.292	10

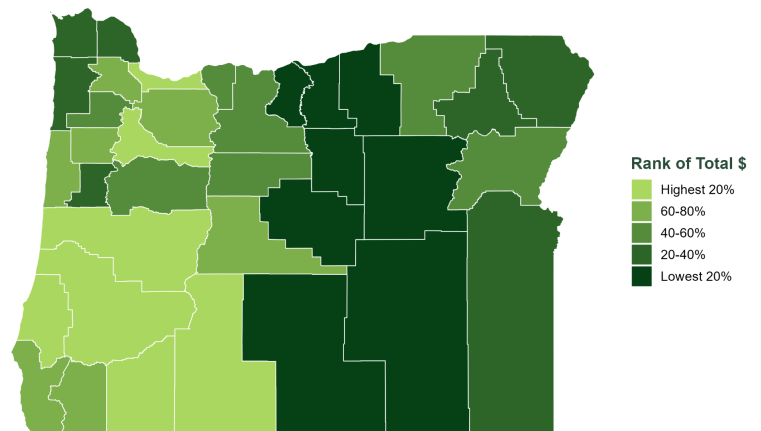
County Rank by Composite Need Index (2021)

Based on OBTN county-level indicators



Total Dollars Given by County (2021-2025)

Grants + Scholarships Combined



Source: TFFF grant and scholarship data

Findings: Need indices were calculated using OBTN indicator z-scores to form a comparative rank under the foundation's strategic framework of Strong Starts, Bright Futures, and Thriving Communities. Counties with the highest need were predominantly rural and concentrated in Southern and Eastern Oregon. This shows initial alignment with grant and scholarship funding. Counties with high need index scores are: Wheeler, Coos, and Douglas.

Q: Does funding align with OBTN measures?

The following tables all use the Spearman correlation coefficient, which measures the strength and direction of a monotonic relationship between funding and OBTN indicators.

+1: A perfect monotonic relationship. As one variable increases, the other increases at a steady rate.

0: No relationship at all between the two variables.

-1: A perfect negative monotonic relationship. As one variable increases, the other decreases.

Correlations Among All Variables

Scholarships			
Variable	Correlation	Need Direction	Strength of Correlation
Largest Community - Population	0.798	N/A (neutral)	Strong
Vehicle Miles Traveled	-0.586	N/A (neutral)	Moderate-Strong
Broadband Access	0.568	↓ Greater Need	Moderate
Transit Service	0.567	↓ Greater Need	Moderate
Kindergarten Readiness: Letter Sounds	-0.561	↓ Greater Need	Moderate
Rent Costs	0.539	↑ Greater Need	Moderate
Mobile Homes	-0.482	N/A (neutral)	Moderate
Child Care Slots	0.413	↓ Greater Need	Moderate
5-yr HS Grad Rate	-0.403	↓ Greater Need	Weak-Moderate
Net Migration	0.402	N/A (neutral)	Weak-Moderate

Scholarships: Scholarship funding and recipients are strongly correlated with urban areas with higher income and university degrees. This suggests a need for further investigation into whether scholarships are going to rural, poorer communities. There is, however, a strong signal that scholarships are supporting communities with low graduation rates and kindergarten readiness scores. This aligns with the foundation's mission to encourage better educational outcomes where students may be struggling, and provide those communities with opportunity for secondary education.

Grants			
Variable	Correlation	Need Direction	Strength of Correlation
Kindergarten Readiness: Letter Sounds	-0.621	↓ Greater Need	Strong
Total Population	0.549	N/A (neutral)	Moderate
Largest Community - Population	0.542	N/A (neutral)	Moderate
Foster Care	0.539	↑ Greater Need	Moderate
Life Expectancy - Total	-0.537	↓ Greater Need	Moderate
Voter Participation	-0.532	↓ Greater Need	Moderate
Rural Population	-0.465	N/A (neutral)	Moderate
Property Tax per Person	-0.462	↓ Greater Need	Moderate
5-yr HS Grad Rate	-0.398	↓ Greater Need	Weak-Moderate
Vehicle Miles Traveled	-0.385	N/A (neutral)	Weak-Moderate

Grants: Grants are highly correlated with kindergarten readiness scores, foster care, and life expectancy. There are moderate correlations with property tax per person and graduation rates. This signals greater alignment between grants and community needs than scholarships. Foster care and kindergarten readiness scores specifically match TFFF’s internal priority of “Strong Starts” and “Bright Futures.” Similar to scholarship correlations, these results merit a deeper look into how dollars are reaching rural communities.

Overall Funding			
Variable	Correlation	Need Direction	Strength of Correlation
Largest Community - Population	0.798	N/A (neutral)	Strong
Vehicle Miles Traveled	-0.586	N/A (neutral)	Moderate-Strong
Broadband Access	0.568	↓ Greater Need	Moderate
Transit Service	0.567	↓ Greater Need	Moderate
Kindergarten Readiness: Letter Sounds	-0.561	↓ Greater Need	Moderate
Rent Costs	0.539	↑ Greater Need	Moderate
Mobile Homes	-0.482	N/A (neutral)	Moderate
Child Care Slots	0.413	↓ Greater Need	Moderate
5-yr HS Grad Rate	-0.403	↓ Greater Need	Moderate
Net Migration	0.402	N/A (neutral)	Moderate

Overall: These findings reflect much of the scholarships influence: the highest correlations stem from urban areas. Foster care and graduation rates remain strong signals, which show some alignment with TFFF strategic goals.

Correlations Among Specified Variables

Need Indicator (Variable)	Need Direction	Scholarship	Grants	Total
3rd Grade Reading	↓ Greater Need	-0.059	-0.306	-0.166
Ninth Grade on Track	↓ Greater Need	-0.012	-0.304	-0.157
5-yr HS Grad Rate	↓ Greater Need	-0.360	-0.398	-0.403
4yr Degree+	↓ Greater Need	0.411	0.049	0.311
Kindergarten Readiness: Letter Sounds	↓ Greater Need	-0.476	-0.621	-0.561
Median Income	↓ Greater Need	0.519	0.004	0.332
Unemployment Rate	↑ Greater Need	-0.284	0.262	-0.080
Property Tax per Person	↓ Greater Need	-0.108	-0.462	-0.248

% Households in Financial Hardship	↑ Greater Need	-0.310	0.084	-0.140
Child Poverty	↑ Greater Need	-0.003	0.317	0.113
Foster Care	↑ Greater Need	0.032	0.539	0.287
Low Weight Births	↑ Greater Need	0.046	0.121	0.064
2yr Old Vaccination Rate	↓ Greater Need	0.114	-0.057	-0.012
Child Care Slots	↓ Greater Need	0.440	0.231	0.413
Food Insecurity	↑ Greater Need	-0.343	0.174	-0.133
Broadband Access	↓ Greater Need	0.653	0.296	0.568

Findings: These comparisons highlight a few key points: firstly, grant funding is much more closely correlated with OBTN need indicators than scholarship funds, which influences the total funding correlations. Secondly, two strong signals continue to emerge through total funding correlations: kindergarten readiness and graduation rates. Foster care, third grade reading scores, and ninth grade on-track scores also show that places with more familial and educational needs are correlated with higher overall funding amounts.

Correlation between need rank & total dollars, grants, scholarships

Category	Correlation
Total Dollars	0.13
Grant Dollars	0.35
Scholarship Dollars	-0.08

Findings: This supports the previous finding that grants are more highly correlated with need. Scholarships have a very weak negative correlation, which skews the total funding results. Even so, total dollars and grants have positive correlations with need index scores: funding is correlated with community need.

Notes: Correlation values rounded to 3 decimal places from original Spearman correlation output (*analysis_script.Rmd*). Need Direction reflects whether a higher or lower raw value of the variable itself signals greater community need, per the project's established *need_direction* metadata (*data_prep.R / ingestion_script.Rmd final_vars* table). This is independent of the sign/direction of its correlation with funding shown in the Correlation column. "N/A (neutral)" variables (e.g. Total Population, Largest Community - Population, Rural Population, Vehicle Miles Traveled, Mobile Homes, Net Migration) are demographic/geographic context measures that the project's methodology does not treat as directional need indicators.

Variable Selection

LASSO LOOCV Method

Variables Selected
5-yr Graduation
Kindergarten Readiness
% Households in Financial Hardship
Foster Care
Rural population

LASSO Bootstrap Method

Variables Selected
5-yr Graduation
Foster Care
Rural population

The table on the left shows variables selected through LASSO L1 regularization using leave-one-out cross-validation on one run of the full dataset, using optimal lambda 1SE. The table on the right shows variables selected through a 1000-repeat LASSO bootstrap with 10-fold CV.

Findings: Core drivers selected by LASSO show themes of education and family (graduation, kindergarten readiness, financial hardship, and foster care indicators). Repeating the LASSO process with a bootstrapping method demonstrates three variables reliably held under significant replication.

Poisson Model on Selected Variables - Results

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    14.4607    0.2173   66.559 < 2e-16 ***
`5-yr HS Grad Rate` -0.3806    0.1260   -3.022 0.004913 **
`Rural Population` -0.8256    0.2145   -3.849 0.000534 ***
`Foster Care`     0.3956    0.1185    3.339 0.002145 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasipoisson family taken to be 1968837)

    Null deviance: 119099307  on 35  degrees of freedom
Residual deviance: 52805593  on 32  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 5

Estimated Dispersion Parameter: 1968837

```

	Log_Beta <dbl>	Multiplier_IRR <dbl>	Pct_Change_Per_Unit <dbl>	Lower_CI_95 <dbl>	Upper_CI_95 <dbl>	P_Value <dbl>
(Intercept)	14.4607	1906265.048	190626404.8	1202380.367	2824690.258	6.872949e-36
`5-yr HS Grad Rate`	-0.3806	0.683	-31.7	0.537	0.887	4.913444e-03
`Rural Population`	-0.8256	0.438	-56.2	0.282	0.654	5.343550e-04
`Foster Care`	0.3956	1.485	48.5	1.180	1.883	2.145406e-03

Findings:

- 5-yr HS Grad Rate: each 1-unit increase is associated with a 31.7% decrease in expected dollars (IRR 0.683, 95% CI 0.537–0.887, $p=0.005$) – significant.
- Rural Population: each 1-unit increase is associated with a 56.2% decrease in expected dollars (IRR 0.438, CI 0.282–0.654, $p<0.001$) – significant, the largest effect.
- Foster Care: each 1-unit increase is associated with a 48.5% increase in expected dollars (IRR 1.485, CI 1.180–1.883, $p=0.002$) – significant.

Gamma Model on Selected Variables - Results

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    14.6732    0.1242  118.131 < 2e-16 ***
`5-yr HS Grad Rate` -0.1619    0.0973   -1.663  0.105984
`Rural Population` -0.5417    0.1264   -4.287  0.000156 ***
`Foster Care`    0.3370    0.1079    3.124  0.003775 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Gamma family taken to be 0.4910908)

    Null deviance: 30.202  on 35  degrees of freedom
Residual deviance: 13.260  on 32  degrees of freedom
AIC: 1135

Number of Fisher Scoring iterations: 11

Estimated Dispersion Parameter (Gamma model): 0.4910908
Warning: glm.fit: algorithm did not converge
```

	Log_Beta <dbl>	Multiplier_IRR <dbl>	Pct_Change_Per_Unit <dbl>	Lower_CI_95 <dbl>	Upper_CI_95 <dbl>	P_Value <dbl>
(Intercept)	14.6732	2357685.220	235768422.0	1870963.288	3033270.506	7.894498e-44
`5-yr HS Grad Rate`	-0.1619	0.851	-14.9	0.652	1.042	1.059843e-01
`Rural Population`	-0.5417	0.582	-41.8	0.463	0.736	1.555286e-04
`Foster Care`	0.3370	1.401	40.1	1.110	1.762	3.774881e-03

Findings: Gamma model has better dispersion and model fit. Unit-change values maintain direction but shrink to 15%, 42%, and 40%, respectively.

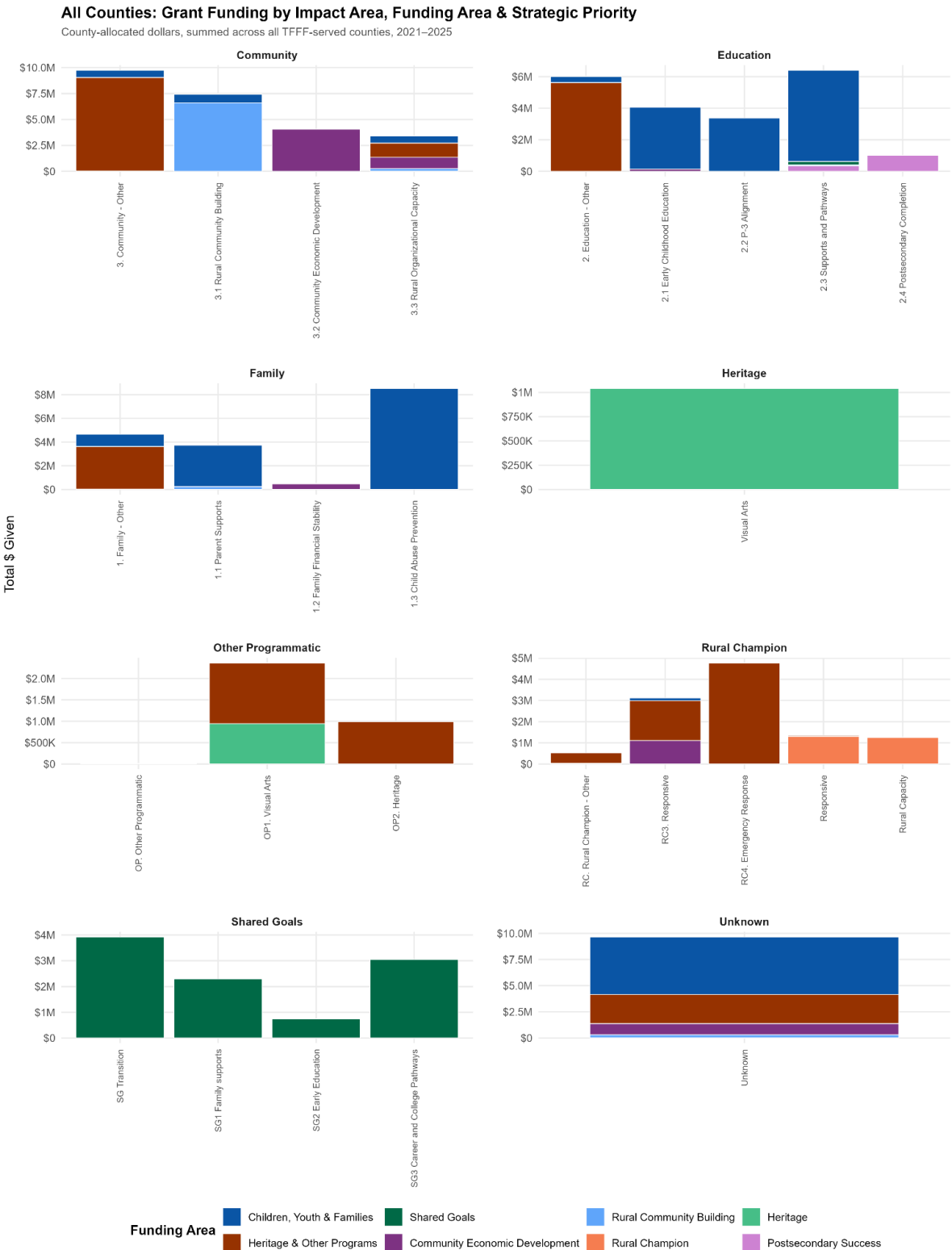
Overall findings:

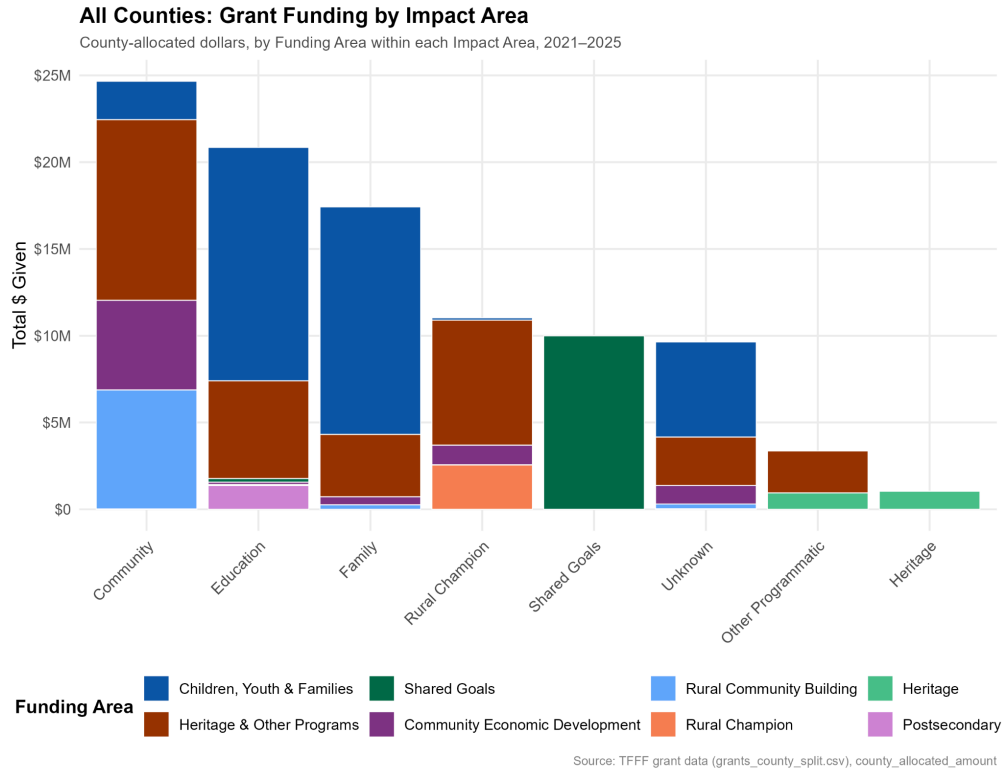
Across the two models, rural population and foster care are statistically significant. In the Poisson model, graduation rates are significant as well.

Deviance measures model fit, with lower values meaning a better fit. Null deviance reflects how poorly an intercept-only model fits the data, while residual deviance reflects fit once the three predictors are added (Bobbitt [12]). In the quasipoisson model, deviance dropped from 119,099,307 to 52,805,593, roughly a 56% reduction, and in the Gamma model it dropped from 30.202 to 13.260, a similar proportional improvement. These diagnostics indicate the predictors meaningfully improve

fit over the null model in both cases. That said, residual deviance in the Quasi-Poisson model remains very large relative to its 32 degrees of freedom, which lines up with the extreme overdispersion already flagged in the model and suggests notable (~44%) unexplained variation remains after accounting for HS Grad Rate, Rural Population, and Foster Care. Those three variables account for over 50% of variation in data, providing notable insight into what is influencing TFFF funding.

Q: Does funding align with strategic goals?



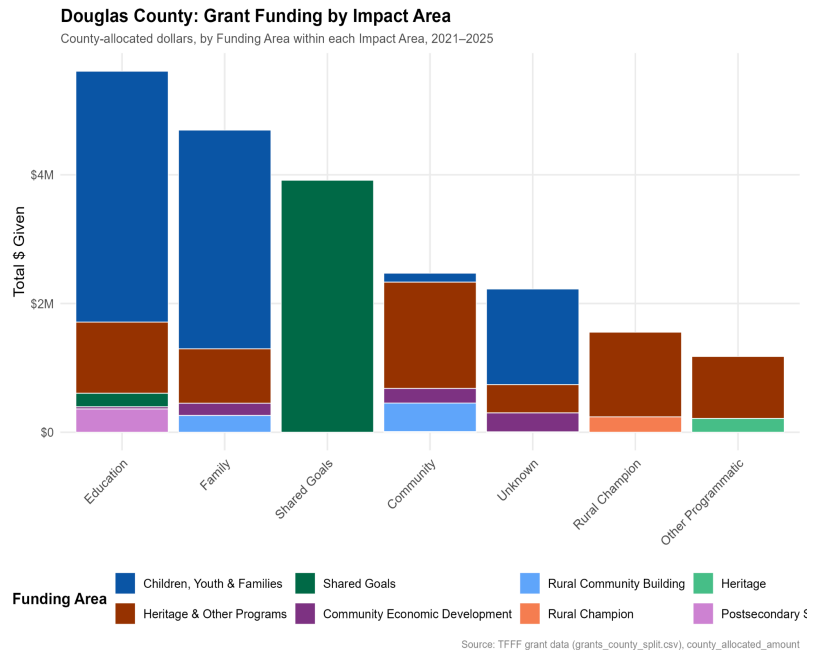


Findings: *Impact Area.* Family, Education, and Community are top priorities for the Foundation. On a statewide level, TFFF invested more in Community than any other category during the 2021–2025 strategic plan. ***Strategic Priority.*** The top funded priorities under this strategic plan were child abuse programs, educational supports and pathways, and rural community building – each counting towards a key TFFF pillar: Strong Starts, Bright Futures, and Thriving Communities.

9.2 Douglas County Profile

Total Funding

Metric <chr>	Amount <chr>
total_scholarship_dollars	\$2,650,914
n_scholarships	127
median_scholarship	\$13,668
total_grant_dollars	\$21,648,387
n_grants	657
median_grant	\$8,000
total_dollars_given	\$24,299,301



Top Grant Recipients

Organization <chr>	Organization Type <chr>	Dollars <chr>
Douglas Education Service District	Education	\$2,881,105
Umpqua Community College	Education	\$1,845,079
The Campbell Institute	Education	\$925,000
Umpqua Community College Foundation	Education	\$715,500
Mercy Foundation	Foundation (Peer Group)	\$658,110

Top need areas in Douglas County (2021)

Indicator	Category	Pillar	Raw Value	State Average	Need Z-Score	Rank
Foster Care	Family	Strong Starts	20.5	10.2	2.134	3
5-yr HS Grad Rate	Education	Bright Futures	71.8	81.6	1.330	3
4yr Degree+	Education	Bright Futures	10.8	15.3	1.293	4
Median Income	Finance	Thriving Communities	47267.0	53953.1	1.105	10
Good Physical Health	Health	Strong Starts	58.0	59.7	1.061	13
Food Insecurity	Family	Strong Starts	14.7	13.1	1.055	9
Child Poverty	Family	Strong Starts	22.1	18.1	0.997	12
LFPR	Finance	Thriving Communities	51.1	56.0	0.796	9
Child Care Slots	Family	Strong Starts	10.0	11.9	0.795	11
Kindergarten Readiness	Education	Bright Futures	6.2	7.9	0.730	12
Adult Tobacco Use	Health	Strong Starts	31.5	29.3	0.676	15
Life Expectancy	Health	Strong Starts	77.3	79.3	0.673	5
Index Crime	Community	Thriving Communities	28.2	21.1	0.614	8
Good Mental Health	Health	Strong Starts	56.5	58.5	0.439	13
Low Weight Births	Health	Strong Starts	7.8	6.3	0.284	8

How have need indicators improved and worsened since the last strategic planning session in Douglas County? (Comparatively to other Oregon counties)

What has improved?

Variable	Category	Shift	Rank Shift	Methodology Changed
Foster Care	Family	-1.201	-5	FALSE

Broadband Access	Geographic	-0.798	-19	FALSE
Low Weight Births	Health	-0.786	-5	FALSE
Index Crime	Community	-0.370	-5	FALSE
Ninth Grade on Track	Education	-0.271	-3	FALSE

What has worsened?

Variable	Category	Shift	Rank Shift	Methodology Changed
Households in Financial Hardship - % HHs in Financial Hardship	Finance	0.734	10	TRUE
Unemployment Rate	Finance	0.540	3	FALSE
2yr Old Vaccination Rate	Health	0.435	6	FALSE
Adult Tobacco Use	Health	0.199	7	FALSE
5-yr HS Grad Rate	Education	0.192	-1	FALSE

Findings: At the county level, funding aligns with need more closely than it does at the state level. In Douglas County specifically, funding shows a heavy focus on education, which matches need reflected in high school graduation rates and four-year degree attainment. Furthermore, investment in family services aligns with foster care and child poverty indicators. Together, these patterns suggest that the majority of documented need corresponds to the "Strong Starts" and "Bright Futures" strategic pillars. This is consistent with funding being weighted more heavily toward education and family than community-level initiatives in Douglas County. One area warranting particular attention is the increase in financial hardship and unemployment in Douglas County between 2021 and 2025, both of which rose more sharply than in other Oregon counties over the same period. This trend, along with worsening health-related indicators (tobacco use and two-year vaccination rates), points to community health as a potential focus for the next strategic plan. As community-focused funding ranked below family and education efforts under the previous strategic plan, these updated 2025 need metrics support the expansion of community-level investment in upcoming years. That said, the magnitude of the financial hardship increases should be interpreted with some caution, as they may partly reflect changes in OBTN's measurement methodology over this period rather than a change in underlying conditions alone.

10 Discussion, Limitations, and Future Work

Interpretation

Overall, total funding given to Oregon and Siskiyou counties from 2021-2025 shows a weak positive correlation with community need, and that relationship is driven mostly by grants. Scholarships

correlate weakly or negatively with need once examined separately. LASSO variable selection identified 5-year graduation, kindergarten readiness, poverty, foster care, rural population, and crime as the clearest predictors of total funding. Of these, foster care, rural population, and graduation rates held up under 1,000-sample bootstrap validation, together accounting for over 50% of the variation in funding across counties. This gives reasonable confidence that these three factors are, in fact, influential.

Successes

Funding aligns well with two of TFFF's three strategic pillars. Dollars track closely with foster care rates (largely driven by the Protect Our Children initiative), consistent with “Strong Starts,” and with graduation and kindergarten readiness scores, consistent with “Bright Futures”. Funds reach communities with the greatest educational and familial need. The overall need-index correlation is positive, meaning funding is trending in the right direction even if the relationship is still weak, and grant funding in particular is well-distributed across all three strategic pillars statewide.

Opportunities

Two gaps stand out. First, funding is heavily concentrated in Douglas County; while responsive grants reach rural, high-need counties in Southern Oregon, Eastern Oregon remains comparatively underserved. The need index points to Wheeler, Coos, Malheur, and Josephine as the counties with the highest unmet need that aren't already receiving Douglas-level investment, something worth prioritizing in the next strategic plan. Second, scholarships correlate with urban population proxies rather than need, and show a negative correlation with the need index overall. They appear less internally goal-aligned than grants, and this gap between the rural priority and actual scholarship distribution merits a closer look.

Douglas County

Locally, funding aligns with need more tightly than it does statewide. Education and family investment track closely with graduation rates, four-year degree attainment, foster care, and child poverty. Post-pandemic Douglas County saw larger increases in financial hardship, unemployment, and health indicators (tobacco use, vaccination rates) than most other counties in Oregon, even as foster care rates improved. That shift suggests the county's needs may be moving toward community and health-related support, which should be considered when partnering with organizations and targeting investments under the next strategic plan.

Limitations

Several limitations apply to the data and design choices made in this project. Missing values were concentrated disproportionately in the state's most rural counties. These values were imputed using a population-peer fallback, and a handful of scholarship zip codes required manual correction – both demonstrate a need for tighter data-entry validation going forward. This missingness pattern is itself a structural concern: county-level aggregation can mask small/rural-county suppression in the underlying OBTN data, and because rural counties have the highest missingness, the analysis risks understating disparities.

Two intentional simplifications shape the results: grant dollars were split evenly across listed impacted counties rather than tied to exact beneficiary organizations, and funding was not normalized by population so as not to mask true rural under-investment. Both are intentional, reasonable choices for this analysis, but worth keeping in mind when interpreting county-level figures. Because several OBTN indicators changed measurement methodologies across editions, valid time-series analysis wasn't possible. The statistical modeling instead relies on a single-year, 36-county sample, which precluded train/test validation and limits how much precision should be read into the model. Douglas County's financial hardship measure specifically changed methodology in 2025 and should be interpreted cautiously when looking at need index shifts. Finally, issues with model fit demonstrated a need for further investigation before being used to inform final funding decisions. The Gamma model returned a convergence warning, and the quasi-Poisson model's overdispersion makes its ~56% pseudo-R² directional rather than precise. While models do not guarantee exact results, they still provide useful information on the strength and direction of the relationship between need indicators and total funding.

Future Work

For the Foundation, next steps should include improved data validation practices and expansion of this analysis. To perform a more complete analysis, zip codes and impacted counties should exist in validated, required fields upon grant and scholarship data entry. Additionally, a better tracking system that follows grant dollars to specific counties, cities, and organizations impacted would allow more accurate results. Trend analysis could be enabled by sourcing more complete time-series data (e.g., Census ACS) to supplement OBTN metrics. Cross-referencing the need index against complementary measures like the CDC Social Vulnerability Index or National Equity Atlas ahead of the next strategic plan would provide further context into the needs of rural Oregon counties. On the analytical side, initial results should be compared to scholarship totals normalized by population to measure how results shift without the rural-priority focus. Future iterations could also test alternative models (OLS, negative binomial) for comparison and incorporate the demographic (race, age, sex) variables, which are already cleaned but not yet used in this round.

11 Conclusions and Recommendations

The Ford Family Foundation's funding under the 2021-2025 strategic plan was aligned with community needs, although there is room for improvement. The top strategic priorities that correlated with actual funded dollars were "Bright Futures" (graduation and literacy rates) and "Strong Starts" (foster care rates). There was also moderate evidence that dollars were given to counties experiencing greater financial hardship (higher poverty rates), consistent with "Thriving Communities." Overall, funding under the previous plan aligned with community needs primarily in the areas of education and family support.

However, results indicate that counties with higher rural populations received less funding. Separating grants from scholarships for further investigation revealed that scholarship dollars and recipient counts predominantly went to more urban areas, which skewed the overall funding results. Values were intentionally left unnormalized for population size, so as not to mask underfunded rural counties. In reference to scholarships, these results may reflect population size rather than funding

priority, as urban counties simply have more residents and therefore more scholarship recipients. Given the strategic priority of expanding opportunity in rural areas, however, this pattern warrants further investigation into scholarship distribution. Quantile-based mapping of grants and scholarships showed substantial funding reaching rural counties in Southern Oregon, while identifying room for growth in Eastern and Northern Oregon. I recommend expanding scholarship outreach specifically in Eastern Oregon's rural communities, both to ensure students are aware these opportunities exist and to give school counselors the support needed to help students apply and interview successfully. Having gone through this process myself, I know firsthand how demanding it can be, and how much difference dedicated support makes. TFFF does this well in Douglas County and the Willamette Valley; it is worth examining whether Eastern Oregon counties receive the same level of support, making this a good place to start as the next area for expansion.

Moving forward, I recommend using the need index to compare county-level needs and guide future strategic priorities. In practice, this means using the dashboard and need index to (1) identify counties for investment in the next strategic plan based on high need paired with low current funding, and (2) identify the top need areas and OBTN indicators within each county to determine which specific initiatives and organizations could be invested in for the greatest impact. One specific recommendation is to shift focus in Douglas County from foster care and child support toward broader community health support, as foster care rates have declined while poverty, unemployment, and health indicators (tobacco use, two-year vaccination rates) have worsened since the previous planning session. While investment in family and education remains strong, expanding funding for community-wide efforts could help increase employment, encourage healthy habits, and improve equitable access to medical and other resources for communities facing financial hardship. This analysis and pipeline should be rerun under the next strategic plan to track improvement and measure alignment over time.

12 Reproducibility

Pipeline

```
data/raw/*.xlsx → ingestion_script.Rmd →  
data/processed/ + data/external/  
▼  
analysis_script.Rmd (stats/models, section 8-11)  
data_prep.R → src/shiny/data/*.rds  
▼  
TFFF_dashboard.Rmd (Run Document)
```

`data_prep.R` reads straight from ingestion output, not from `analysis_script.Rmd`. The dashboard and the modeling results are parallel paths.

Data Access

TFFF's source data is confidential under the executed Data Use and Confidentiality Agreement and is not distributed with this repo. OBTN is public (tfff.org/oregon-numbers). Grants and Scholarship files were provided directly by TFFF and aren't otherwise available. Reproducing this pipeline requires requesting access from TFFF directly. `docs/data-dictionary.md` should list source, URL, and approval status per file (Approved as of 6/17/2026).

Output is split by sensitivity, not just by stage: `data/raw/` and `data/external/` (org names, dollar-level detail, recipient PII: `grants.csv`, `grants_county_split.csv`, `scholarship.csv`, `recipients_ready.csv`) are gitignored; `data/processed/` (county-level aggregates only: `obtn_clean.csv`, `race_long.csv`, `age_sex_long.csv`) is tracked.

Steps

- 1. Ingestion:** Place the 4 raw workbooks in `data/raw/`, update the path variables at the top of `ingestion_script.Rmd`, run top to bottom.
- 2. Analysis:** Update the `read_csv()` paths in `analysis_script.Rmd` to match Step 1's output; run top to bottom. Produces the correlation tables, need index, and quasi-Poisson/Gamma models in sections 8-11.
- 3. Dashboard:** In `src/shiny/`: update paths in `data_prep.R` (including `path_grants`, `path_scholarship`, `path_recipient`) and run it; then open `TFFF_dashboard.Rmd` and click Run Document.

Dependencies & Limitations

Key packages: tidyverse, stringr, readxl, zipcodeR, sf, tigris, glmnet, flexdashboard, shiny, leaflet, scales, plotly, GGally, car, purrr, sf, tigris (Census shapefiles) and zipcodeR (zip database) are external, non-TFFF dependencies requiring their own first-run download.

All paths are hardcoded to one Windows machine; every script needs manual editing to run elsewhere. LASSO/bootstrap use `set.seed(42)`, but exact reproduction depends on matching glmnet versions.

13 References

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